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Axial behaviour of underground steel pipeline buried in unsaturated soils

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ABSTRACT

Soils surrounding underground pipelines are often unsaturated in field conditions. However, the effects of the unsaturated condition on the axial interaction between pipelines and soils are unclear and not considered in current design guidelines. This limitation may lead to unsafe designs, as the load from potentially moving soil to pipelines could be underestimated. To address this, six large-scale physical modeling tests were conducted to examine pipe pullout behavior using a rough steel pipe buried in saturated and unsaturated completely decomposed granite (CDG). Matric suctions around the pipe (0 ~ 70.2 kPa) were adjusted by varying the initial water content and measured using tensiometers. The results show that axial pullout resistance, under constant nominal overburden pressure, increases with suction. At a suction of 70.2 kPa, the resistance was 1.69 times greater than in the saturated condition, highlighting significant risks in current design guidelines. 68 % of this axial resistance increment is attributed to the additional interface contact pressure induced by capillary forces of soil-pipe interface liquid menisci. The remaining 32 % is related to net interface contact pressure increases mainly due to suction effects on constrained dilatancy. A new and simple model was developed for calculating axial resistance in the unsaturated condition, based on elastic expanding cylinder theory, considering suction effects on Bishop's stress, stiffness, and dilatancy.

Notation

$A(\gamma)$, $M(\gamma)$	strain-dependent empirical parameters related to G and G_u
a, n, m	fitting parameters in the van Genuchten model
c_{int}, c_{int}'	interface shear strength intercept and effective interface shear strength intercept if assuming a linear failure envelope
δ, δ'	interface friction angle and effective interface friction angle if assuming a linear failure envelope
D	pipe outer diameter
d_{10}	particle size at which 10 % of the soil sample's mass is finer
d_{50}	average particle size
e	void ratio
G, G_u	shear modulus for saturated or dry soils and unsaturated soils
H_c, H_{c0}	buried depth of the pipe center and real buried depth of the pipe center in the experiments
K_0	coefficient of lateral earth pressure at rest
P	air pressure inside the flexible air bag
p_r	reference pressure of 100 kPa
R_{max}, R_a	maximum height and average height deviations on the surface profile over a travel length of d_{50}
R_n	normalized surface roughness based on R_{max}
s	matric suction

(*continued on next column*)

Notation (*continued*)

S_r	degree of saturation
T	pullout resistance, the maximum axial force per meter length
u, u_c	axial displacement and critical axial displacement
u_a, u_w	air pressure and pore pressure
γ	shear strain
γ', $\Delta\gamma$	effective unit weight and unit weight difference between the pipe and backfill soil
$\Delta\sigma_D'$	average increase in interface contact pressure caused by constrained dilation
Δh	buried depth of tensiometer related to that of the pipe springline
ΔW	normal force increase per unit length caused by the pipe weight
κ, κ_{int}	empirical exponent parameter of Bishop's stress for pure soils and soil-interface
σ_c', $\sigma_{c\text{-}net}$	nominal vertical pressure at the pipe center and net nominal vertical pressure at the pipe center
σ_n', $\sigma_{n\text{-}net}$	effective normal stress and net normal stress
$\sigma_{i\text{-}net}$	initial net stress
σ_{int}, σ_{int}'	total interface contact pressure and effective interface contact pressure
$\sigma_{int\text{-}net}$, $\bar{\sigma}_{int\text{-}net}$	net interface contact pressure and the average value of net interface contact pressures

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Notation (*continued*)

τ_f	interface shear strength
φ'	effective internal friction angle of soils
χ, χ_{int}	Bishop's stress parameter for pure soils and soil-interface
ψ_{max}, $\psi_{max\text{-}u}$	maximum dilation angle and maximum dilation angle of unsaturated soils

1. Introduction

Underground pipelines, for various purposes such as water mains and gas and oil lines, have significantly enhanced living standards since the early days of civilization (Almohammed et al., 2018; Moser and Folkman, 2008). Relative displacement between pipelines and surrounding soils in various directions, caused by permanent ground deformations such as landslides and earthquakes, imposes loads on the pipe (Abdelfateh et al., 2021; Fattah et al., 2018b; Karamanos et al., 2021; Lee et al., 2016; Ni and Mangalathu, 2018; O'Rourke et al., 1995; Peng and Yin, 2025; Zhang et al., 2025). The current study focuses on the axial soil-pipeline interaction (ASPI). Typically, pipelines are shallowly buried above the underground water table, where the soil is often partially saturated and exhibits negative water pressure (i.e., matric suction, s) (BSI, 2003; WSD, 2012). The suction effect, which significantly impacts interface shear strength and dilatancy, is a critical factor affecting axial load (Cui and Zhou, 2024; Hamid and Miller, 2009; Hossain and Yin, 2015; Li et al., 2021, 2025; Liu et al., 2021b). Ignoring its influence on axial load evolution can compromise the safety of pipeline systems.

Commonly used design guidelines and previous studies typically analyze the soil-pipe interaction based on the traditional soil mechanics framework that is suitable for dry and saturated soils (Ala, 2001; Alotaibi et al., 2021; Bilgin and Stewart, 2009a,b; Fattah et al., 2018a; Guo and Zhou, 2025a; Murugathasan et al., 2021; O'Rourke et al., 2008; Reza and Dhar, 2021; Scarpelli et al., 2003; Sheil et al., 2018; ASCE, 1984; Huber and Wijewickreme, 2014; O'Rourke and Nordberg, 1992; PRCI, 2009). For instance, ALA (2001) calculates the axial pullout resistance (i.e., the maximum axial force per unit length) of pipes buried in soils, T, using the equation $\pi D[(1/2)(1 + K_0)\sigma_c' \tan \delta']$, where D is the pipe outer diameter; δ' is the interface friction angle. The term $(1/2)(1 + K_0)\sigma_c'$ is the assumed average soil-pipe interface contact pressure, which is the mean of the nominal vertical effective pressure σ_c' ($=\gamma' H_c$) and its lateral earth pressure at rest $K_0\sigma_c'$. Here, γ' is the effective unit weight of the backfill sand and H_c is the buried depth of the pipe center. In this equation, the impact of partial saturation is unclear.

Recently, several experimental and numerical studies have been conducted to enhance the understanding of pipeline behavior in unsaturated soils (Jung et al., 2013; Peng and Yin, 2025; Peng et al., 2024; Randeniya et al., 2020; Robert et al., 2016a; Robert et al., 2016b). For instance, Jung et al. (2013) compared the lateral soil-pipeline interaction in dry and unsaturated sands through physical model tests, finding that the lateral resistances of coarse Cornell sand (d_{10} = 0.216 mm) under saturated and unsaturated conditions were nearly identical. In contrast, Robert et al. (2016b) reported that the lateral resistance of fine Chiba sand (d_{10} = 0.13 mm) under unsaturated conditions was 2.5 times greater than that under dry conditions. As for the soil-pipe interaction in the axial direction, until now, only Al-Khazaali and Vanapalli (2019) conducted large-scale physical modeling to explore the effects of unsaturated conditions on ASPI. They buried a rough steel pipe in clean sand and controlled matric suction by adjusting the water table. They observed an increasing trend in pullout resistance as the average measured matric suction increased from 0 (submerged condition) to 7.25 kPa, followed by a decrease as suction further increased to 9.75 kPa.

In practice, local coarse-grained soils containing a certain amount of fine particles, such as completely decomposed granite (CDG) in Hong Kong (Zhu et al., 2018), are typically used instead of clean sands. Particularly, when pipes are buried on steep slopes, silty or clayey soils are preferred as backfilling materials due to their ease of compaction (USBR, 1996). Compared to clean sand, the suction effects on ASPI using soils with fine particles are expected to be more pronounced with three factors as follows. First, under constant normal loading (CNL) conditions, the suction effect on the interface shear strength of soils with fine particles, mainly due to the additional interface contact pressure due to capillary forces of liquid menisci between the interface and soil particles, is more notable. For example, Borana et al. (2016) observed that the interface shear strength between CDG and rough steel could increase by approximately 40 ~ 70 kPa as suction rises from 0 to 50 kPa. This higher interface shear strength of unsaturated soils with fine particles could lead to a more substantial increase in axial pullout resistance compared to clean sand. Second, various experimental and numerical studies (Guo and Zhou, 2025a; Murugathasan et al., 2021; Wijewickreme et al., 2009) have reported an increase in interface contact pressure due to constrained dilatancy. Unsaturated soils with fine particles exhibit higher dilation and stiffness than saturated soils and unsaturated sand. It leads to more significant interface contact pressure increments and, consequently, increased axial resistance (Alonso et al., 2010; Han and Vanapalli, 2016; Hossain and Yin, 2010b; Salgado et al., 2000). Third, the lateral earth pressure coefficient at rest, K_0, typically decreases with increasing suction (Lu and Likos, 2004; Ng et al., 2024), which might influence the average interface contact pressure on the pipe surface. Neglecting the three factors related to the suction effect on axial load can compromise the resilience of the pipeline system. Specifically, underestimating axial resistance may lead to a lower design strength for the pipe, reducing its ability to withstand soil loads during events such as landslides. Therefore, a new physical modeling study is expected to quantify the axial resistance of pipes buried in soils with fine particles and investigate the suction effects on the above three factors.

This study presents six large-scale physical modeling tests aimed at investigating suction effects on ASPI using a steel pipe buried in CDG, a type of clayey sand, under varying saturation conditions. To the best of the authors' knowledge, this is the first investigation into the axial interaction between pipelines and unsaturated soil containing fine particles, as opposed to clean sand. The primary objectives are: (a) to assess the extent to which matric suction affects axial pullout resistance, (b) to analyze the mechanisms of suction effects on ASPI, and (c) to propose and validate a new model for calculating axial resistance of pipes buried in unsaturated soils.

2. Experimental setup

2.1. Experimental system

This study improved the experimental apparatus and methodology developed by Guo and Zhou (2025a) to assess ASPI in unsaturated conditions (see Fig. 1). It includes a steel box to simulate the pipeline trench, a steel pipe, a servo linear actuator with a reaction frame for axial displacement application, and a flexible air bag for vertical loading. Two sleeves are installed inwards on two openings of the front and rear walls of the steel box to minimize the boundary effects of these walls. The coaxial alignment of the pipe and sleeves resulted in an 8.5 mm gap (109 d_{50} of CDG), allowing for pipe settlement (less than 2 mm in this study) and a soil-pipe interface shear zone (2 to 10 d_{50} (DeJong and Westgate, 2009)). This configuration ensures that the pipe is subjected solely to loads from the soil and maintains a constant soil-pipe interaction length of 0.7 m during pullout. Two flexible rubber membranes were fixed with the pipe and the sleeves, respectively, to prevent soil leakage and provide adequate lateral support for surrounding soils. A connection with free vertical displacement between the pipe and servo linear actuator is shown in Fig. 2. A flexible air bag was positioned between the sample and the top cap. The net nominal vertical pressure at the center, $\sigma_{c\text{-net}}$, of the unsaturated conditions could be regulated by adjusting the air pressure, P, inside the bag, following the equation $\sigma_{c\text{-net}} = \gamma H_c = \gamma H_{c0} + P$. Here, H_{c0} represents the actual burial depth from the

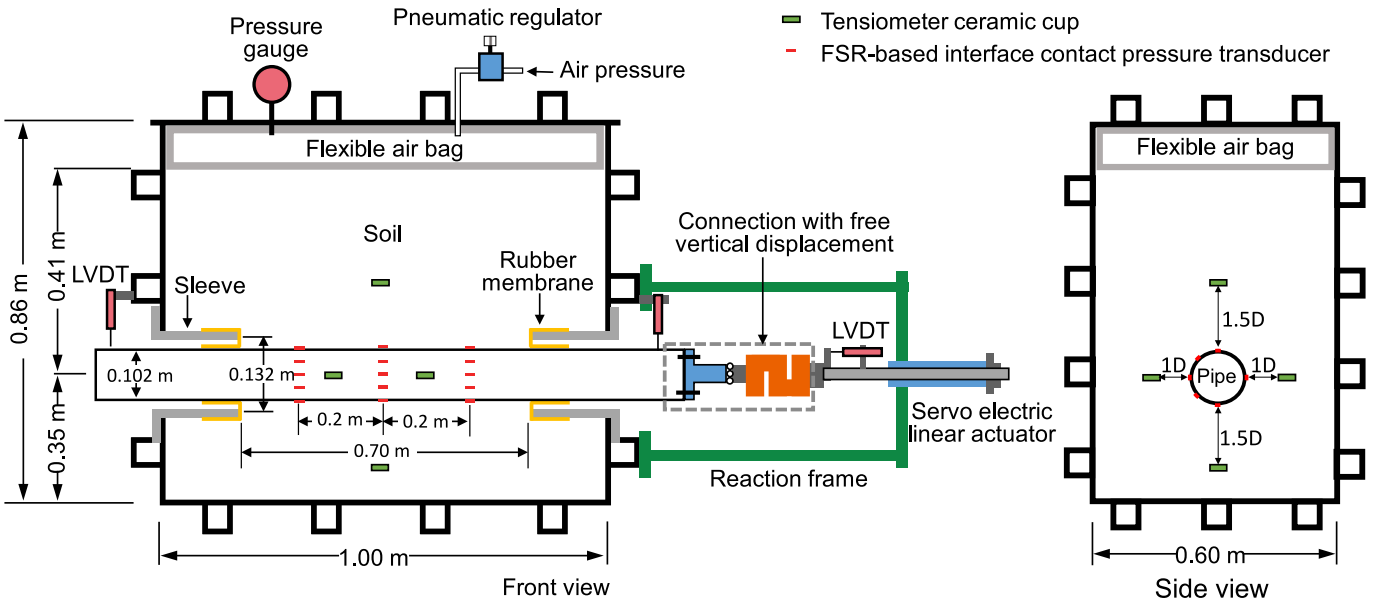


Fig. 1. Schematic diagram of the experimental system for axial soil-pipe interaction physical modelling (after Guo and Zhou (2025a)).

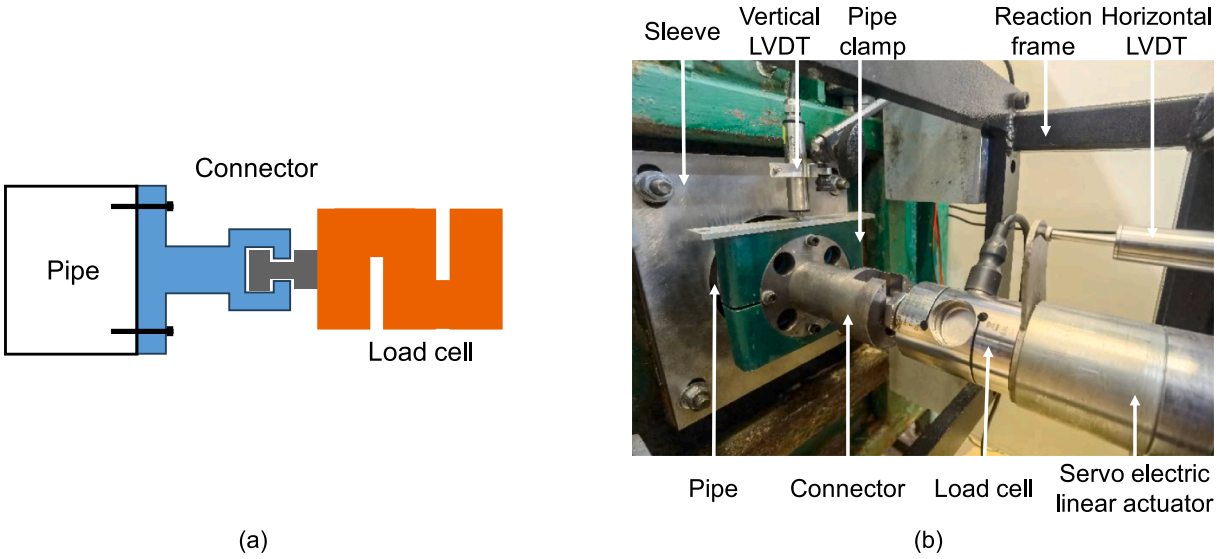


Fig. 2. Pipe-actuator connection with free vertical displacement: (a) schematic diagram of top view; (b) photograph.

ground surface to the pipe's center. As for the saturated condition, the effective nominal vertical pressure at the center, σ_c', would be set as $\gamma' H_{c0} + P$.

‡R_n: Normalised surface roughness using R_{max}/d_{50}.

A seamless steel pipe with a nominal diameter of 102 mm and a nominal thickness of 4 mm, commonly used in Hong Kong, was utilized (BSI, 2002; HKIUS, 2011). The axial resistance per unit area ($T/\pi D$) provided in this study could be a reference for pipes with other diameters or materials. The effects of fluid weight on the pipe pullout mechanism might be insignificant in most cases, and they are not considered here. Further details can be found in Table 1. The surface roughness of the pipe was quantified using the normalized roughness (R_n) proposed by Kishida and Uesugi (1987): $R_n = R_{max}/d_{50}$. Here, R_{max} is the maximum height deviation on the surface profile over a travel length of d_{50}. The pipe was machined using a turning method to achieve R_n values of 1.02 (i.e., R_{max} = 0.08 mm) to simulate surface roughness common in practice.

2.2. Instrumentation

The instrumentation is shown in Fig. 1 and Fig. 2. Two linear variable differential transformers (LVDTs) were set on two pipe ends to measure the pipe vertical displacement. A load cell with ± 50 kN and an LVDT with 50 mm was set between the pipe and the actuator to monitor the axial force and displacement.

A cutting-edge sensor, Force Sensing Resistor (FSR), was utilized to measure soil-pipe interface contact pressure (i.e., total stress) (Interlink, 2023). This piezoresistive sensor's electrical conductance varies as a function of the applied pressure. Its thinness and flexibility are particularly advantageous in adapting to curved surfaces (Kootahi and Leung, 2022; Liu et al., 2021a). The active area of FSR has a diameter of 12.7 mm, which is 163 d_{50} of the soil used in this study. It is larger than the threshold of 10.5 d_{50} suggested by Kootahi and Leung (2022), beyond which the particle size effects on the measurement accuracy are minimal. The installation configuration of the FSR on the pipe surface is depicted in Fig. 3. Holes with a diameter of 8 mm were drilled into the

Table 1
Soil and pipe properties.

Soil Properties	
Unified Soil Classification System (ASTM, 2017)	SC
Specific gravity* (Hossain and Yin, 2010)	2.60
Median particle size, d_{50}: mm	0.078
Plastic limit*: % (Hossain and Yin, 2010)	22.7
Liquid limit*: % (Hossain and Yin, 2010)	32.8
Maximum dry density: kg/m^3 (standard Proctor tests, ASTM (2021))	1780
Optimum moisture content: % (standard Proctor tests, ASTM (2021))	15.1
Target relative compaction: %	95
Effective internal friction angle, φ': ° (Geoguide 1, 2020)	38 ~ 42
Effective cohesion: kPa (Geoguide 1, 2020)	0 ~ 5
Elastic shear modulus: MPa (Wang and Ng, 2005)	50 ~ 200
Poisson's ratio (Du et al., 2021)	0.15 ~ 0.40
Permeability: m/s (Geoguide 1, 2020)	10^{-7} ~ 10^{-6}
Pipe Properties	
Diameter, D: mm	102
Wall thickness: mm	4
Total length: m	1.25
Testing length: m	0.7
Steel density: kg/m^3	7930
Young's modulus: MPa	204
Poisson's ratio	0.3
R_{max}: mm	0.08
$R_n^{\ddagger}$	1.02
R_a: mm	0.313

pipe to allow for wiring. Following the wiring process, the FSRs were affixed to the pipe surface and coated using Epoxy Adhesive DP100 (manufactured by 3 M™). After a curing period of 48 h, the epoxy adhesive formed a rigid coating layer. This coating serves to mitigate errors arising from uneven pressure distribution on the sensor (Interlink, 2023). Additionally, a heat-shrinkable tube was applied at the end of the FSR to provide further protection for the wiring connection. Eighteen FSRs were placed over three cross-sections spaced 0.2 m apart. Each cross-section featured six FSR sensors positioned at the crown, one shoulder, two springlines, one haunch, and the invert. The performance of FSRs is highly dependent on specific installation conditions. Therefore, calibration of the FSRs was conducted after installation. A pipe repair clamp, consisting of a rubber sleeve with thickened edges and a steel split sleeve with screws, was modified to create a sealed space on the pipe surface with the FSRs. During calibration, the air pressure in this space was controlled using a pneumatic regulator, ranging from 0 to 200 kPa, with a loading rate of 4 kPa/min. More details of the sensor can be found in Guo and Zhou (2025a) and Guo and Zhou (2025b).

Six remote tensiometers (2100F), manufactured by Soilmoisture Equipment Corp., with a vacuum dial gauge range, were used to measure the matric suction around the pipe. The nominal measurement range of the tensiometer is 0 to 100 kPa. However, readings exceeding 80 kPa are generally considered unreliable due to the increased risk of cavitation (Ng and Leung, 2012). The vacuum dial gauges were calibrated and adjusted to ensure accurate readings prior to use each time. The distribution of the tensiometers' porous ceramic cups at their buried positions is illustrated in Fig. 1. To maximize the sensitivity, the porous ceramic cups were fully saturated before installation. After initially filling the tensiometer bottle and removing air from the vent tube, the service cap was connected to a vacuum pump set at −20 kPa through a waste bottle. The porous ceramic cup was immersed in a water tank to create seepage over the ceramic. This process lasted for at least 12 h. To prevent damage, the ceramic cups were installed through holes in the box walls. Once the soil layer above the target buried position was compacted, a copper tube with an outer diameter slightly smaller than that of the ceramic cup was inserted through the hole to loosen the soil. Subsequently, the ceramic cup was covered with a layer of saturated clay slurry to ensure good soil-cap contact and to minimize the possibility of air entry. Finally, the ceramic cups were inserted into their target positions and secured using cable glands mounted on the box walls for waterproofing.

2.3. Test materials

The testing soil is completely decomposed granite (CDG), typical in-situ and trench-backfilling soil in Hong Kong (Zhu et al., 2018). The particle size distribution curve and key properties of CDG used in this study are shown in Fig. 4 (a) and Table 1. The plastic and liquid limits of the fines of CDG are 22.7 and 32.8 % as measured by Hossain and Yin (2010b) based on BSI (1990). Based on the Unified Soil Classification System (ASTM, 2017), this soil is classified as clayey sand (SC). The tested CDG is a coarse-grained soil, and it is often modeled using the Mohr-Coulomb in the engineering practices of Hong Kong (Ho et al., 2013; Hossain and Yin, 2010a; Lam et al., 2011). The maximum dry density and optimum water content are 1780 kg/m^3 and 15.1 % using standard Proctor tests (ASTM, 2021). It should be noted that the data provided in this study could serve as a reference for understanding the interaction between pipes and other soils qualitatively. The results of the current study should be interpreted with caution when applied to different soils, pipe materials, or ranges of suction and stress. The target relative compaction is 95 %, corresponding to the target dry density of 1691 kg/m^3, meeting the compaction requirement in practice (USBR, 1996).

Fig. 4 (b) illustrates the soil–water retention curve of CDG measured in the drying path by Hossain and Yin (2010b) using a suction-controllable direct shear apparatus via the axis-translation method under a net normal pressure of 0 kPa. It can be fitted by van Genuchten model (van Genuchten, 1980) as

$$S_r = [1 + (as)^n]^{-m} \quad (1)$$

where a, n, and m are fitting parameters, equaling 0.07, 0.11, and 0.7, respectively, for CDG used in this study.

A series of CNL interface direct shear tests were conducted to evaluate the interface shear behavior between CDG and interfaces treated like those on pipes. The apparatus used for CNL interface direct shear tests is modified from the conventional direct shear test apparatus, as illustrated in Fig. 5. The shear rate was set at 0.02 mm/s, matching the pipe pullout speed used in subsequent physical modeling. The soils used in interface direct shear testing were collected after the following ASPI physical modeling from areas at the same elevation as the pipe springline, but at least 100 mm away from the pipe and tensiometer caps. The

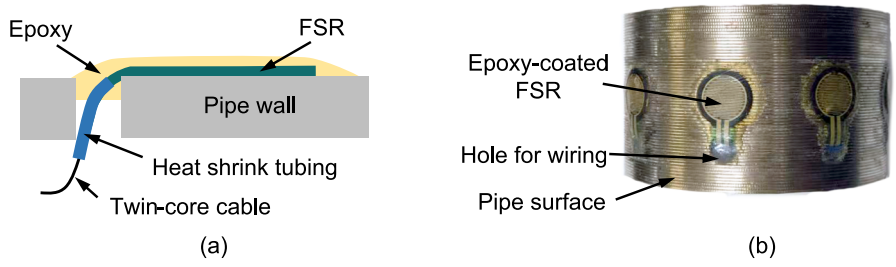


Fig. 3. FSR-based interface contact pressure transducer: (a) schematic diagram of top view; (b) photograph.

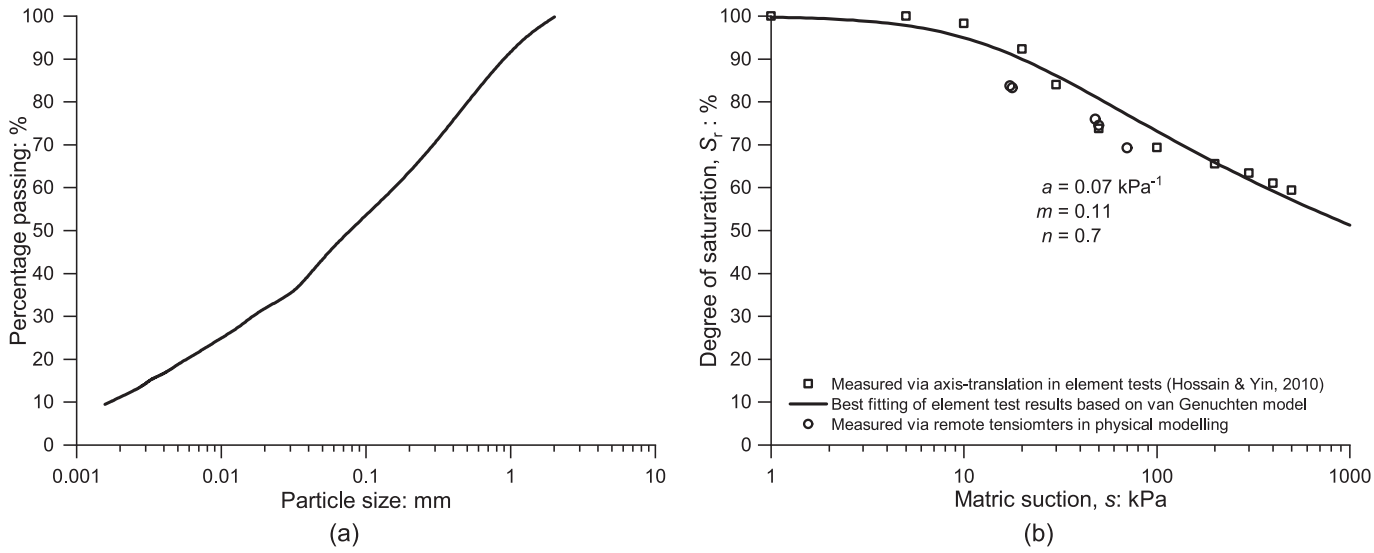


Fig. 4. Properties of completely decomposed granite (CDG): (a) particle size distribution curve; (b) soil–water retention curve.

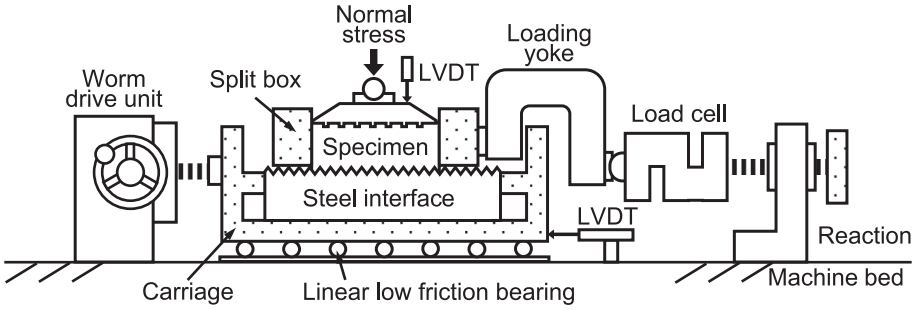


Fig. 5. Schematic diagram of the apparatus for soil-interface direct shear tests.

collected soil was sieved through a 2 mm mesh and remolded into 60 × 60 × 20 mm^3 specimens, maintaining the initial density. This process was conducted swiftly and carefully to ensure that the water content and suction remained consistent. The effective normal stresses applied were 17, 34, 50, and 100 kPa, corresponding to the target nominal pressure at the pipe center in the physical modeling. Fig. 6 illustrates typical interface shear behaviors for both saturated and unsaturated soils. Saturated soils under 17 and 34 kPa exhibited dilation behavior with strain-softening in the stress curves and positive vertical displacements of 0.036 and 0.018 mm, respectively. As the normal stress increased, soils displayed compaction behavior, with vertical displacement reaching −0.15 mm at a surcharge of 100 kPa. For unsaturated CDG with a matric suction of 50.0 kPa, significant dilation behaviors were observed under all normal pressures. It is indicated by pronounced softening in the stress curves and positive vertical displacements. The shear strength results are presented in Table 2. The interface shear strength, τ_f, of unsaturated CDG can be predicted by the Mohr-Coulomb model, as widely used in previous studies by Hossain and Yin (2015), Borana et al. (2016), and Cui et al. (2024), expressed as

$$\tau_f = c'_{int} + (\sigma_{n-net} + \chi_{int} s)\tan\ \delta' \quad (2)$$

where c_{int}' and δ' are the effective interface shear strength intercept and effective interface friction angle with the assumption of a linear failure envelope. $\sigma_{n\text{-}net} + \chi_{int}s$ is the Bishop's interface stress. $\chi_{int}s$ is the additional interface normal stress induced by capillary forces of liquid menisci between the interface and soil particles. χ_{int} is Bishop's interface stress parameter for soil-interface (Bishop, 1959), taken as $S_r^{\kappa int}$, where κ_{int} is an empirical exponent parameter. In this study, κ_{int} is assumed to be a constant value, similar to the empirical exponent parameter κ used for Bishop's stress parameter for pure soils (S_r^{κ}) (Han and Vanapalli, 2016). κ_{int} might be related to the interface roughness and interface material hydrophilicity and therefore is expected to differ from that of the pure soil (Vanapalli et al., 1996). For the CDG and rough steel interface used in this study, the best fit for κ_{int} is 2.5, slightly larger than the typical κ value of 2 for pure clayey and silty soils (Han and Vanapalli, 2016). The Coefficient of Determination (R^2) value for the prediction of shear strength results in Table 2 is as high as 0.99.

2.4. Testing programme

Considering the duration of each test, this study mainly focuses on the matric suction effect by six physical modeling tests, as summarized in Table 3.. The matric suction was adjusted by varying the water content during compaction. Measuring matric suction greater than 80 kPa in physical modeling is challenging. Therefore, the target water contents were 14.5 %, 15 %, and 17 %, corresponding to 70 ~ 82 % in the degree of saturation. Referring to the soil–water retention curve in Fig. 4 (b), matric suctions at these target water contents are below 80 kPa. For comparison, one test was conducted under saturated conditions. The target nominal pressures, $\sigma_{c\text{-}net}$, were set at 17 and 34 kPa, corresponding to burial depths of 1 and 2, respectively.

The values of matric suction measured by tensiometers before the pipe pullout are presented in Table 4. In each test, the distribution of suction values generally aligns with hydrostatic conditions, indicating a limited effect of soil heterogeneity. During the pullout, the variation in suction is typically within ± 1 kPa. This allows the average suction before pullout to be considered constant during shearing. Additionally,

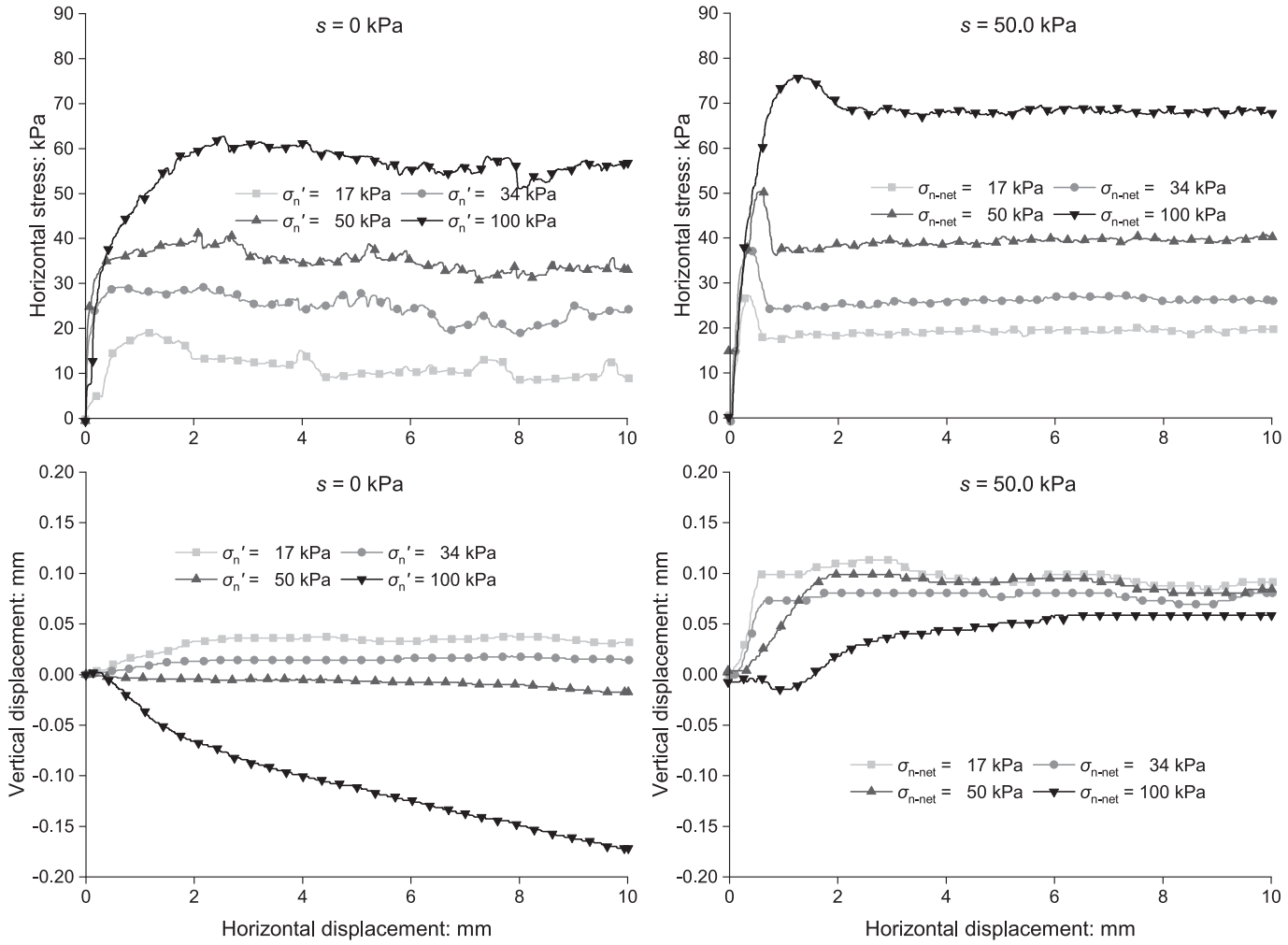


Fig. 6. Unsaturated CDG-interface direct shear tests.

Table 2
Interface shear strength of unsaturated CDG and pipe.

Peak interface shear strength: kPa		Normal stress, $\sigma_{n\text{-net}}$ or σ_n': kPa				Parameters of Mohr-Coulomb model	
Degree of saturation, S_r: %	Matric suction, s: kPa	17	34	50	100	c_{int} or c_{int}': kPa	δ or δ': °
100	0	19.0	29.2	41.2	62.2	12.0	27.2
83.3*	17.3*	26.0	32.6	46.6	70.5	16.5	28.7
74.5*	50.0*	27.4	37.5	50.3	76.0	18.4	30.3
69.2*	70.2*	28.1	40.1	53.7	79.7	19.5	31.6

*Data from Table 4.

Table 3
Testing programme.

Reference	Target water content: %	Target degree of saturation: %	Nominal pressure at the center, $\sigma_{c\text{-net}}$ or σ_c': kPa
P17Wsat	20.7 (saturated)	100	17
P17W17	17.0	82.3	17
P34W17	17.0	82.3	34
P17W15	15.0	72.5	17
P34W15	15.0	72.5	34
P17W14	14.2	68.3	17

the actual water content was determined by randomly collecting five soil samples, each weighing at least 40 g, from areas at the same elevation as the pipe springline after tests. The water content at the pipe springline elevation is slightly higher than the target water content used during mixing. This discrepancy is attributed to the redistribution of water content during hydrostatic equilibrium. Fig. 4 (b) illustrates the relationship between the measured matric suction and water content, represented by the degree of saturation. These data points are slightly lower than those obtained using the axis-translation method in element tests. This difference is primarily because the compaction process resembles a wetting path rather than the drying path in element tests, where the soil retains less water in the former path.

2.5. Model preparation and test procedures

CDG, adjusted to the target water content, was thoroughly mixed in a blender and crumbled to pass through a 600 × 400 mm^2 sieve with 10 × 10 mm^2 meshes. The soil was then compacted using a 1900 W electric earth rammer equipped with a 150 × 150 mm^2 plate to achieve the target high relative compaction, similar to engineering practices in the field (USBR, 1996; WSD, 2012). The models were constructed in layers, each approximately 50 mm thick, to meet the practical requirement of 150 mm (USBR, 1996). During sample preparation, sleeves were installed at an elevation of 0.28 m. At 0.33 m, a pipe with rubber membranes was placed in the box. The soil around the sleeve ends was temporarily removed. After positioning the membranes, the soil was backfilled and compacted to its original state using a wooden mallet.

Table 4
Measurement of matric suction and water content around the pipe before pullout.

Reference	Matric suction before pullout, s: kPa							Water content after testing†: %	Saturation after testing, S_r: %
	$\Delta h^* = 0.2\ m$	$\Delta h^* = 0\ m$				$\Delta h^* = -0.2\ m$	Average		
P17Wsat	–	–	–	–	–	–	0	20.0‡	96.8
P17W17	21	17	17	18	19	15	17.8	17.2	83.3
P34W17	19	18	17	17	18	15	17.3	17.3	83.7
P17W15	52	52	48	51	50	47	50.0	15.4	74.5
P34W15	50	45	51	51	42	47	48.0	15.7	76.0
P17W14	75	74	67	68	73	64	70.2	14.1	69.2

*Δh: Buried depth of tensiometer related to that of the pipe springline.
†Measurement by five samples close to the pipe springline.
‡Estimation based on the total volume of added water.

CDG was then continuously added and compacted until the elevation reached 0.76 m.

The saturated sample was initially compacted with a target water content of 17 %. Water was injected from the bottom of the sample through four holes in the box walls, using a water cylinder with an elevation of 2.0 ~ 2.4 m relative to the box bottom. This process was maintained for seven days to maximize venting through seepage. A plate with a small air vent covered the box to minimize evaporation. The saturation was roughly estimated at 96.8 % based on the total volume of added water.

After sample preparation, the flexible air bag and top cap were placed on the soil, with the air pressure of the bag adjusted to the target value. This condition was maintained for at least seven days until the vertical position of the pipe and the values of matrix suction stabilized. Subsequently, the pipe's axial velocity was set at 0.02 mm/s, with a target axial displacement of 20 mm.

3. Experimental results and discussion

3.1. Axial behavior of steel pipe buried in saturated and unsaturated soils

Fig. 7 (a) illustrates the axial displacement-force relationship across all tests. The results indicate that pipes in both saturated and unsaturated soils exhibit clear strain-softening behavior. This observation is consistent with the results of soil-interface direct shear tests, which demonstrate interface dilatant behavior, as shown in Fig. 6. Under a nominal pressure of 17 kPa, the axial pullout resistance increases from 4.44 to 11.95 kN/m as the average suction rises from 0 to 70.2 kPa. The ratio of axial resistance in unsaturated soil to that in saturated soil reaches as high as 2.69. It underscores the crucial need to account for suction effects on axial resistance during pipeline design. Failing to do so could result in the pipe experiencing stress that exceeds its design strength, potentially leading to damage when undergoing any axial displacement related to the surrounding soil.

Fig. 7 (b) presents the vertical displacement behavior for all tests. A clear and continuous settlement is observed in the curve for the saturated condition, with the settlement eventually reaching 1.35 mm. In contrast, the settlement under unsaturated conditions is not as evident. The maximum settlements for matric suctions ranging from 17.2 to 50.0 kPa do not exceed 0.22 mm (2.8 times d_{50}). The test with an average suction of 70.2 kPa exhibits uplifting behavior rather than settlement. These vertical displacement curves reflect the influence of soil-pipe interface behavior due to two distinct mechanisms. On one hand, soil compression of the soil beneath the pipe due to variations in average effective stress tends to cause the pipe to settle. On the other hand, the dilation behavior of the soil-pipe interface shear band below the pipe tends to lift the pipe. The stronger compression modulus of unsaturated CDG compared to saturated soil reduces the impact of the first mechanism, while its enhanced interface dilatancy leads to more pronounced pipe lifting. Consequently, the settlement of the pipe in unsaturated CDG is less than that in saturated CDG, and there is even lifting when the suction reaches 70.2 kPa.

3.2. Effects of suction and buried pressure on pullout resistance

Fig. 8 illustrates the relationship between axial pullout resistance per unit area, $T/(\pi D)$, and the average matric suction around the pipe. The increase in the interface shear strength due to the additional interface

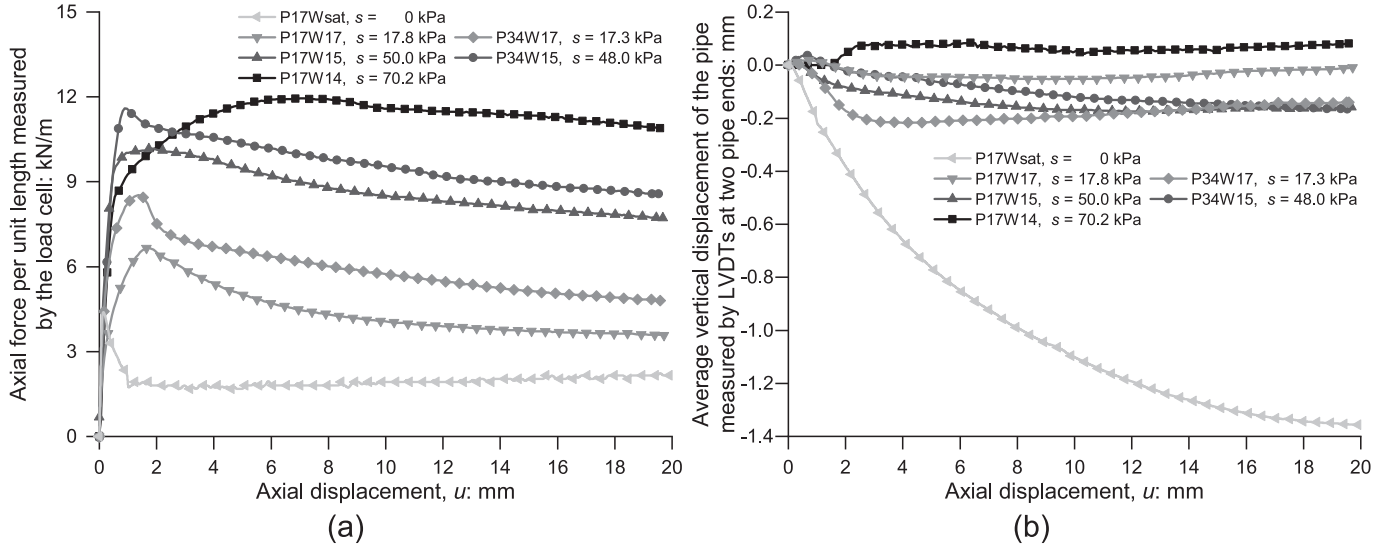


Fig. 7. Axial behavior of pipe: (a) axial force against axial displacement; (b) vertical displacement against axial displacement.

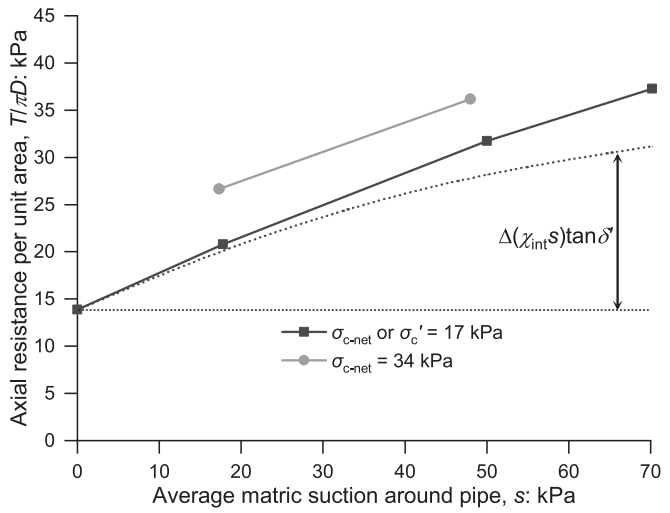


Fig. 8. Axial resistance against average matric suction.

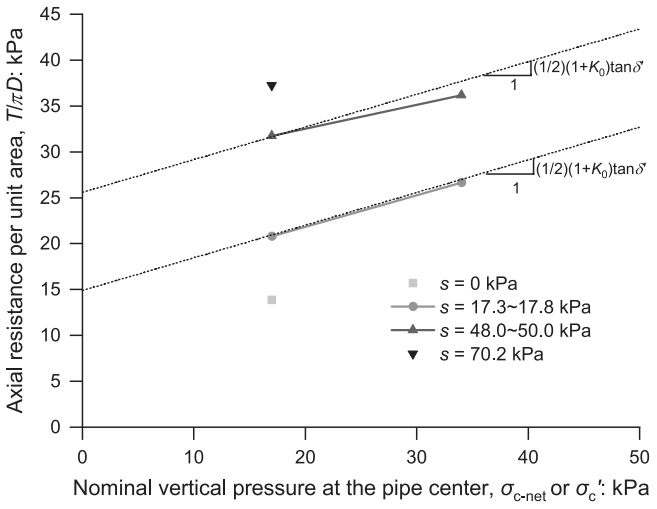


Fig. 9. Axial resistance against nominal vertical pressure at the pipe center.

contact pressure induced by capillary forces ($\Delta(\chi_{int}s)\tan\delta'$, according to Eq. (2)) is included for comparative analysis. As matric suction rises from 0 to 70.2 kPa, axial resistance per unit area increase ($\Delta T/(\pi D)$) under $\sigma_{c\text{-net}}$ of 17 kPa is up to 25.4 kPa, while the value of $\Delta(\chi_{int}s)\tan\delta'$ is 17.3 kPa. This indicates that around 68 % of $\Delta T/(\pi D)$ (i.e., 17.3/25.4) is due to the increase in Bishop's interface contact pressure. As for the remaining 32 %, it mainly results from the stronger constrained dilation behavior in unsaturated soils. The presence of soil away from the interface constrains the dilation trend at the interface, leading to increased net interface contact pressures during shearing. Constrained dilation-induced interface contact pressure increase is a function of soil dilatancy and stiffness (Luo et al., 2000), which is discussed later. The dilatancy of the unsaturated CDG-pipe interface is positively correlated with matric suction when suction is below at least 300 kPa (Hossain and Yin, 2015). Moreover, the stiffness of soil with fine particles is also found to generally increase with suction (Han and Vanapalli, 2016). Therefore, the difference between $\Delta T/(\pi D)$ and $\Delta(\chi_{int}s)\tan\delta'$ increases from 10 to 32 % of $\Delta T/(\pi D)$ with suction increasing from 17.8 to 70.2 kPa. Furthermore, it should be acknowledged that axial resistance may not exhibit a continuous increase with further increments in matric suction. A principal limitation of the present study lies in the fact that the maximum matric suction was restricted by the limitations of the matric suction measurement technique. Consequently, uncertainties remain regarding the behavior under field conditions characterized by higher matric suctions, particularly in arid regions. Future investigations could address these uncertainties by employing advanced numerical simulation methods or by enhancing matric suction measurement techniques in large-scale physical model tests.

Fig. 9 demonstrates that axial pullout resistance increases with the nominal pressure at the pipe center. The slopes of $T/(\pi D)$ as net nominal pressure increases from 17 to 34 kPa for pipes subjected to suctions of 17.3 ~ 17.8 and 48.0 ~ 50.0 kPa are 0.343 and 0.261, respectively. These values are slightly smaller than $(1/2)(1 + K_0)\tan\delta'$ (0.356), the theoretical slope based on the assumption by ALA (2001). K_0 is typically taken as 1-sinφ', where φ' is taken as 38° in this study. This trend may be attributed to another interface contact pressure variation mechanism, where matric suction reduces the lateral earth pressure at rest.

3.3. Suction effects on interface contact pressures distribution and evolution

Fig. 10 presents the distribution and evolution of interface contact pressures for tests conducted under nominal vertical pressure of 17 kPa, with average suction ranging from 0 to 50.0 kPa. The interface contact pressure measured by FSR, σ_{int}, is the total stress. For saturated soils, the effective interface contact pressure σ_{int}' (=σ_{int}-u_w) is used below. Here, u_w is pore pressure, which exhibits limited variation during the pipe pullout process. Hence, u_w is taken as hydrostatic pressure in this study. For unsaturated soils, the interface contact pressure can be treated as net interface contact pressure $\sigma_{int\text{-net}}$ (=σ_{int}-u_a), where u_a (= 0 kPA in this study) is the air pressure. The variation in interface contact pressures measured by the FSR sensors is generally within ± 16 % of the average value at corresponding positions across the three cross-sections. The difference in contact pressure between the left and right springlines remains below 2.8 kPa, indicating good symmetry. Consequently, the average values at the crown, shoulders, springlines, haunches, and invert across three cross-sections are presented directly. According to guidelines, the interface contact pressures should be σ_c' or $\sigma_{c\text{-net}}$ (17 kPa) at the crown and invert, $K_0\sigma_c'$ or $K_0\sigma_{c\text{-net}}$ (6.53 kPa) at the springlines, and $(1/2)(1 + K_0)\sigma_c'$ or $(1/2)(1 + K_0)\sigma_{c\text{-net}}$ (11.77 kPa) at the shoulders and haunches.

Before the pullout of the pipe buried in saturated soils, all interface contact pressures, except at the invert, are slightly higher than the expected values. This discrepancy may be attributed to the high level of compaction, causing stress concentration across the pipe surface. This is a phenomenon frequently observed in previous physical model tests (Guo and Zhou, 2025a; Sheil et al., 2018). During the pullout process, the interface contact pressure at the pipe invert increases by approximately 4 kPa at an axial displacement of 0.4 mm, corresponding to the peak axial force in Fig. 7 (a) and the settlement behavior in Fig. 7 (b). Changes in interface contact pressure at other locations are minimal, aligning with the limited interface dilatancy of saturated CDG shown in Fig. 6.

Before pipe pullout in unsaturated soils, net interface contact pressures at the springline are 4.18 and 7.59 kPa for matric suctions of 17.8 and 50.0 kPa. The average initial net interface contact pressures at matric suction of 0, 17.8, and 50.0 kPa are 13.59, 10.21, and 11.93 kPa, respectively. These values deviate by no more than 15.5 % from the classical empirical formula $K_0 = 1 - \sin\varphi'$, which doesn't consider the suction effect. During the pullout process, interface contact pressures at different locations generally increase. The increase at the pipe invert is most pronounced early in the pullout. They reach maximums of 22.7 and 48.3 kPa for average matric suction of 17.8 and 50.0 kPa, respectively, playing a crucial role in axial pullout resistance. Interface contact pressures at the springline for matric suction of 17.8 and 50.0 kPa decrease to 4.06 and 3.82 kPa, respectively, at the axial displacement when axial force peaks. This results from the equilibrium of initial concentration on the pipe interface disturbed during pullout, causing lateral earth pressure to approach the theoretical lateral earth pressure at rest. Late-stage pullout interface contact pressure increases

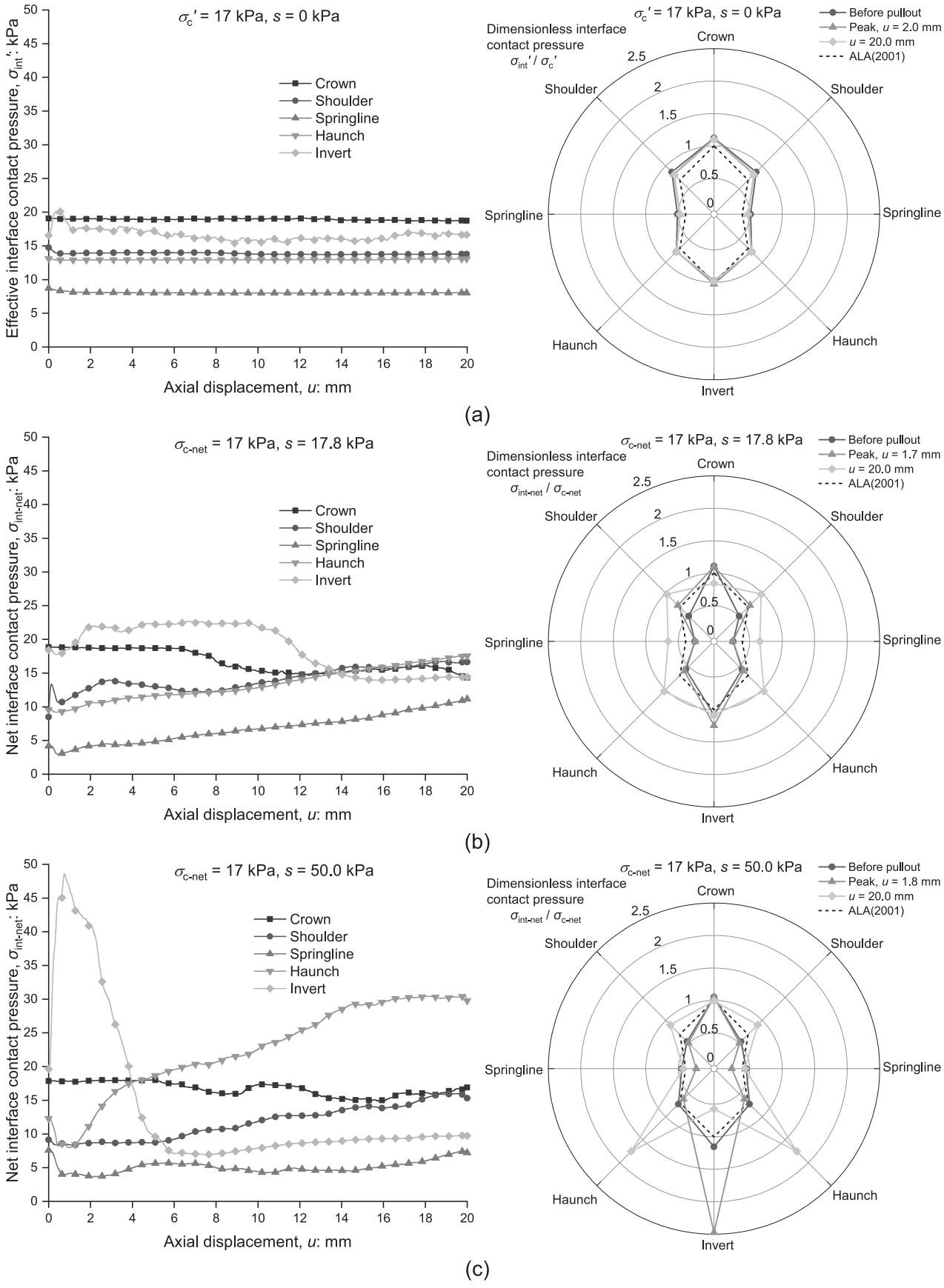


Fig. 10. Interface contact pressure development and distribution ($\sigma_{c\text{-net}}$ or σ_c'= 17 kPa): (a) $s = 0$ kPa; (b) $s = 17.8$ kPa; (c) $s = 50.0$ kPa.

concentrate on the shoulder and haunch, reaching 12.50 and 10.13 kPa for suction of 17.8 kPa and 8.65 and 10.44 kPa for suction of 50.0 kPa. The overall interface contact pressure increase is more significant for tests with higher average matric suction. It is due to stronger constrained dilation with increasing suction, supporting the above finding where the difference $\Delta T/(\pi D)$ and $\Delta(\chi_{int}s)\tan\delta'$ increases with suction in Fig. 8.

Moreover, the redistribution of interface contact pressure caused by the soil arching effect (e.g., the higher pressure at the pipe crown and haunches than the predicted values) is less pronounced than in previous tests using sands by Sheil et al. (2018) and Guo and Zhou (2025a). This may be because the higher compression modulus of saturated and unsaturated CDG minimizes the soil arching effect on the pipe.

3.4. Suction effects on stress path

Fig. 11 illustrates the stress paths at the soil-pipe interfaces in physical modeling. The horizontal axis represents Bishop's interface contact pressure as $\overline{\sigma}_{int\text{-}net} + \chi_{int}s$, where $\overline{\sigma}_{int\text{-}net}$ is the average value of net interface contact pressures. The interface failure envelopes of saturated CDG, obtained through CNL interface direct shear tests, are included as references. The shapes of the stress paths are influenced by the average matric suction. In saturated conditions, due to limited variation in interface contact pressure, the stress path rises vertically towards the peak failure envelope and then descends vertically back to the critical state failure envelope.

In unsaturated conditions, the evolution of the stress path becomes more complex. For the matric suction of 17.8 kPa, the path includes three stages, resembling the typical stress path observed in CNS tests on the rough interface of dry sands (Ooi and Carter, 1987; Pra-ai and Boulon, 2016) and in the physical modeling tests on the rough pipe buried in dry pure sand by Guo and Zhou (2025a). In the first stage, the stress path rises vertically, indicating that friction angle mobilization is dominated by elasticity. The path shape during this stage is flexural due to the dynamic equilibrium between the increment in interface contact pressure caused by constrained dilation and the decrease in lateral earth pressure caused by shear disturbance as in Fig. 10. In the second stage, the path moves towards the top right, approaching the peak failure envelope, illustrating that the increment in interface contact pressure due to constrained dilation has become dominant. Finally, the path moves to the bottom right and stops near the critical state envelope, indicating a transition of the friction angle from peak to critical state and further constrained dilation behavior. For the suction of 50.0 kPa, the stress path during the first and second stages is similar to that of 17.8 kPa, initially rising vertically and then turning to the top right. The major difference occurs before the final stage, where the path moves to the bottom right; here, the stress path shifts towards the bottom left initially. This shift is due to the non-synchronized evolution of interface contact pressures at different surface locations, as shown in Fig. 10 (a), where the interface contact pressure at the invert rises sharply to 48.3 kPa and then descends sharply back to 7.0 kPa.

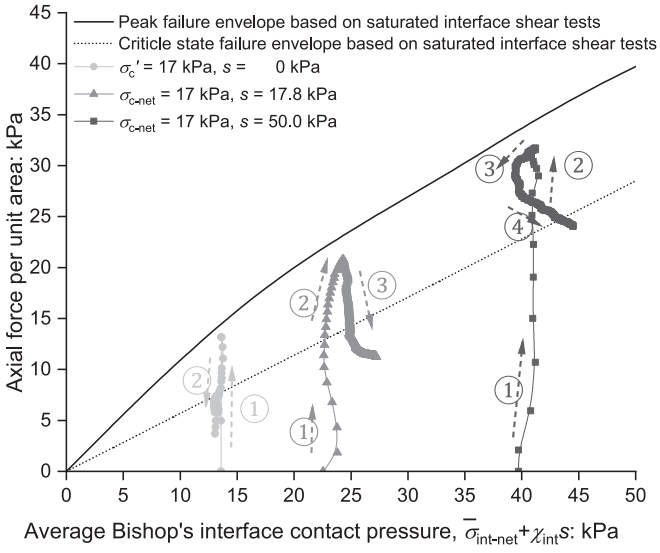


Fig. 11. Stress path based on average interface contact pressure measured by FSR.

4. A new method for calculating the pullout resistance of pipe in unsaturated soils

4.1. Development of the new method

Guo and Zhou (2025a) proposed a simple equation that accounts for constrained dilation and the self-weight of the pipe to calculate pullout resistance for pipes buried in dry sand:

$$T = \pi D\left(\gamma' H_c \frac{1+K_0}{2} + \Delta\sigma_D' + \Delta W/\pi D\right)\tan\ \delta' \tag{3}$$

where $\Delta\sigma_D'$ represents the average increment in interface contact pressure due to constrained dilation, and ΔW (= $(1/4)\pi D^2\Delta\gamma$) is the normal force increment per unit length caused by the pipe's weight. Here, $(1/4)\pi D^2$ is the pipe's volume per unit length; and $\Delta\gamma$ is the difference between the apparent unit weight of the pipe and the bulk unit weight of the soil.

The calculation on axial pullout resistance of pipes buried in unsaturated soils can be extended by combining Eq. (3) with Eq. (2) as

$$T = \pi D\left[c_{int}' + \left(\frac{1+K_0}{2}\sigma_{c-net} + \chi_{int}s + \Delta\sigma_D' + \Delta W/\pi D\right)\tan\ \delta'\right] \tag{4}$$

Given the lack of a reliable formula for K_0 in unsaturated soils, applying the classical empirical formula (i.e., $K_0 = 1 - \sin\varphi'$, which doesn't consider the suction effect) to estimate the average initial net interface contact pressure is a practical approach. When suction formulations are available, they can be readily incorporated into the proposed model. $\Delta\sigma_D'$ varies with suction. It can be determined by the extended elastic expanding cylinder theory by Luo et al. (2000) as

$$\Delta\sigma_D' = 2G\frac{u_c}{D}\tan\ \psi_{max} \tag{5}$$

where u_c is the critical axial displacement where the mobilized dilation angle reaches its peak value; ψ_{max} is the maximum dilation angle of the soil; and G is the soil shear modulus. To simplify the complex interface behavior associated with dilation and stiffness, two key assumptions are incorporated into the expanding cylinder theory. First, the tangent of the mobilized dilation angle is assumed to increase with axial displacement until it reaches its maximum value, ψ_{max}. Second, it is assumed that both the mobilized dilation angle and the mobilized pullout resistance reach their peak values at the same axial displacement, u_c. More details are available in Guo and Zhou (2025a). The method of Audibert and Nyman (1977) is used to determine u_c from Fig. 7 (a). This involves drawing a horizontal line through the maximum axial force and a secant line from the origin to the point of 70 % of the maximum axial force. Their intersection determines u_c. Table 5 summarises the values of u_c in this study and Al-Khazaali and Vanapalli (2019), ranging from 0.09 to 2.85 mm. G and ψ_{max} depend on matric suction and degree of saturation, typically being larger in unsaturated soils than in saturated or dry soils (Dai and Zhou, 2025; Han and Vanapalli, 2016; Hossain and Yin, 2015). The shear modulus of unsaturated soils, G_u, can be determined by replacing effective stress as Bishop's stress in the shear modulus equation for dry or saturated soils by Guo and Zhou (2025a) as the method of Sawangsuriya et al. (2009), expressed as

Table 5
Parameters for new equation verification.

Parameters	Present study						Al-Khazaali and Vanapalli (2019)				
Nominal pressure, $\sigma_{c\text{-net}}$ or σ_c': kPa	17	17	17	17	34	34	4.42*	4.22*	3.47*	3.35*	3.25*
Average suction, s: kPa	0	17.8	50.0	70.2	17.8	48.0	2.75	4.5	7.25	8	9.75
Unit weight difference, $\Delta\gamma$: kN/m^3	0.05	0.62	0.92	1.10	0.05	0.07	0.22	1.21	4.95	5.55	6.06
Critical displacement, u_c: mm	0.09	0.80	0.39	0.74	0.44	0.55	0.53	1.39	0.20	0.37	0.64
Maximum dilation angle, ψ_{max}: °	0	5.71	5.71	5.71	5.71	5.71	6.75	6.75	6.75	6.75	6.75
Pipe diameter, D: mm	102						114.3				
Interface friction angle, δ': °	27.2						34.5				
Interface shear strength intercept, c_{int}': kPa	12.0						0				
Exponent parameter of Bishop's stress for the interface, κ_{int}	2.5						0.1†				
Effective internal friction angle, φ': °	38						35.3				
Void ratio, e	0.538						0.606				
Exponent parameter of Bishop's stress for pure soils, κ	2‡						1‡				
Shear modulus, G: kPa	$A(\gamma) = 6.113$, $M(\gamma) = 0.448$						$A(\gamma) = 126$, $M(\gamma) = 1$				

*Estimation based on average suction and soil–water retention curve.
†Back-fitting value based on physical modeling and recommended by Al-Khazaali and Vanapalli (2019).
‡Recommended by Han and Vanapalli (2016).

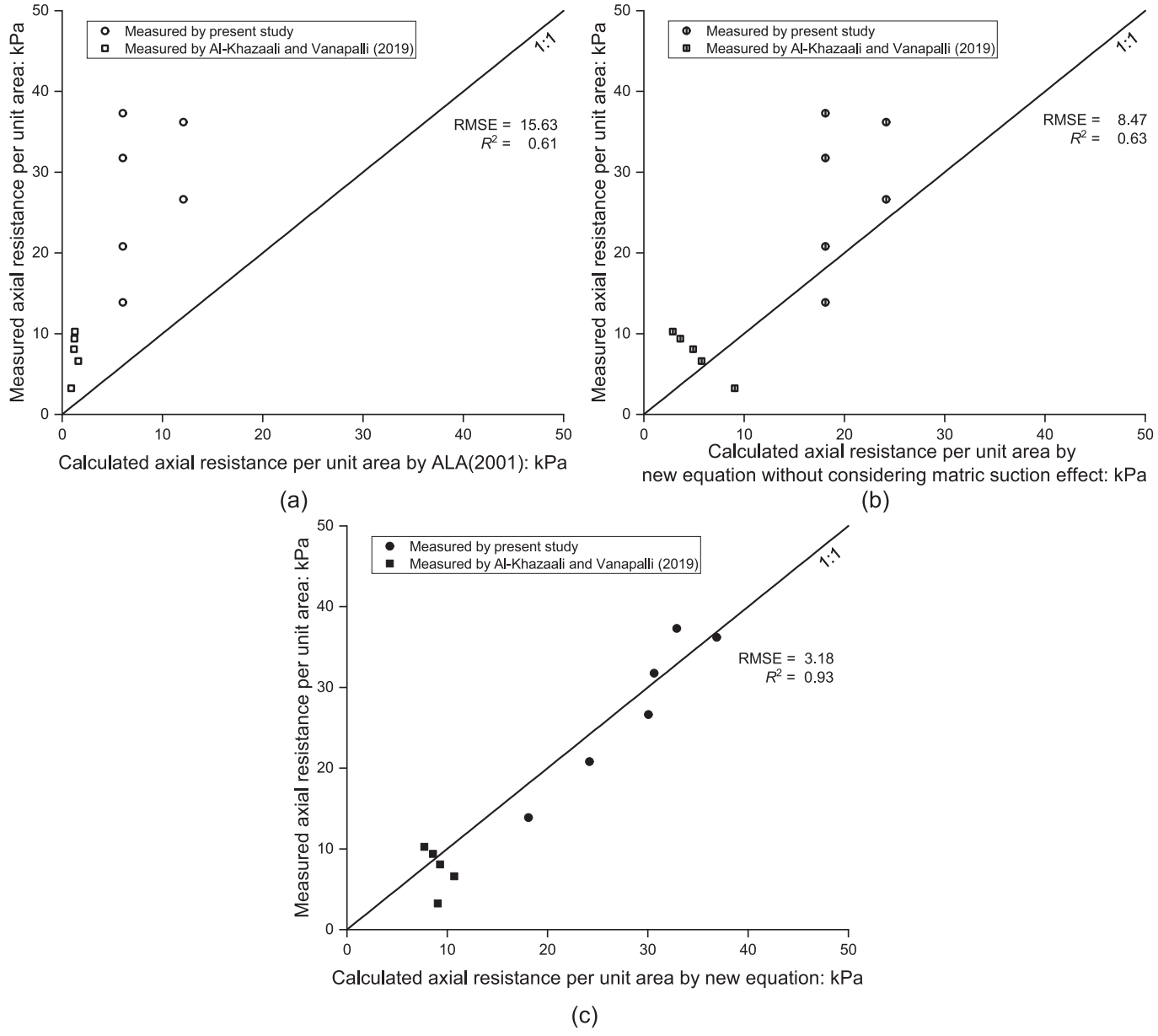


Fig. 12. Axial pullout resistance calculated by (a) ALA (2001); (b) new equation without considering the matric suction effect; (c) new equation.

$$G_u/p_r = \frac{A(\gamma)}{(1+e)^3}\left(\frac{\sigma_{i-net}+\chi s}{p_r}\right)^{M(\gamma)} \quad (6)$$

where $A(\gamma)$ and $M(\gamma)$ are the material parameter and the effective stress exponent parameter, which are functions of the shear strain γ (i.e., the ratio of axial displacement to shear band thickness). p_r is the reference pressure of 100 kPa. $\sigma_{i\text{-net}}$ is the initial net stress, taken as $\sigma_{c\text{-net}}(1+K_0)/2$ in this study. χ $(=S_r^{\kappa})$ is Bishop's stress parameter for pure soils **(**Bishop, 1959**)**, where κ is an empirical exponent parameter and κ value is typically assumed to have a constant value of 2 for clayey and silty soils (e.g., CDG) and 1 for cohesionless soils (Han and Vanapalli, 2016). Based on the values of u_c in Table 5 and the empirical interface shear band thickness (DeJong and Westgate, 2009), the critical interface shear strains at which the axial force reaches their peak values are determined to be no less than 10 %. For CDG, based on a regression analysis of 12 bender element measurements reported by Wang and Ng (2005), the values of $A(\gamma)$ and $M(\gamma)$ for $\gamma = 10$ % are determined as 6.113 and 0.448. As for clean sand, such as Unimin 7030 quartz silica sand used by Al-Khazaali and Vanapalli (2019), Oztoprak and Bolton (2013) developed a model of G based on a database of 454 tests from the literature, where the empirical values of $A(\gamma)$ and $M(\gamma)$ are 126 and 1 for $\gamma = 10$ %.

Secondly, ψ_{max} in Eq. (5) is normally higher in unsaturated soil than that of dry or saturated soil, especially for the silty and clayey soils. However, there are limited models to predict ψ_{max} of unsaturated soil ($\psi_{max\text{-}u}$). For CDG, values of ψ_{max} and $\psi_{max\text{-}u}$ are determined as 0° and 5.71°, respectively, based on the direct shear tests of CDG by Hossain and Yin (2010b). As for the sand of Al-Khazaali and Vanapalli (2019), a constant value of 6.75° from their direct shear test of dry soil is used for both saturated and unsaturated conditions, considering the lack of data and minimal suction effect on $\psi_{max\text{-}u}$ of pure sand. Once a reliable model for the maximum dilation angle of unsaturated soil is developed, it is expected to be integrated into the proposed equation.

In summary, the new equation of unsaturated soil can be expressed as

$$T = \pi D\left[c'_{int} + \left(\frac{1+K_0}{2}\sigma_{c-net} + \chi_{int}s + 2G_u\frac{u_c}{D}\tan\ \psi_{max-u} + \frac{D}{4}\Delta\gamma\right)\tan\ \delta'\right] \quad (7)$$

Table 5 summarises the parameters for the new simple equation, including data from Al-Khazaali and Vanapalli (2019) and the present study. Fig. 12 shows the calculated results based on these parameters. To evaluate the performance of the new equation, its results are compared with those of ALA (2001) and with the new equation applied without considering the matric suction effect (i.e., excluding the $\chi_{int}s$ term and using the values of G and ψ_{max} under saturated conditions). The Root Mean Square Error (RMSE) and R^2 of the new model are 3.18 and 0.93, respectively, demonstrating significantly better performance compared to the values of 15.63 and 0.61 for the prediction of ALA (2001) and 8.47 and 0.63 for the calculated results without considering matric suction.

4.2. Recommendations for updating pipeline design

In the field, pipelines are often subjected to complex and multi-directional loads resulting from permanent ground deformations. In current engineering analysis and design (ALA, 2001; ASCE, 1984; PRCI, 2009), soils surrounding pipelines are often modeled as several groups of discrete linearly elastic-perfectly plastic Winkler springs in the axial, vertical, and lateral directions within finite element method (FEM) simulations. Ground deformations can be simulated by adjusting the base positions of these springs. Further details and illustrative examples can be found in ALA (2001), PRCI (2009), and Xie et al. (2011).

An important input parameter for the above method is resistance (i.e., the maximum force per unit length) in each loading direction. The results of the current study may offer valuable insights into axial resistance. To account for the effects of matric suction on ASPI, it is recommended that Eq. (7) be employed. The parameters related to Bishop's stress, shear strength, dilatancy, and stiffness in Eq. (7), as well as the soil–water retention curve, can be determined through element tests conducted on the target soil and pipe materials. The potential range of matric suction is influenced by weather conditions, groundwater tables, and other factors and can be determined through field monitoring or numerical simulations. Given the possible range of matric suctions, the maximum and minimum values of axial resistance can be calculated, based on which a proper design value can be used. It should be emphasized that, in different scenarios, both underestimation and overestimation of axial resistance can lead to unsafe designs. When analyzing the response of a pipeline to moving soils, underestimating the axial resistance may result in a lower design value for pipeline strength, thereby increasing the risk of pipeline rupture. Conversely, when analyzing the probability of buckling failures (e.g., upheaval buckling or sharp bend ruptures) during heating or other actions, overestimating the axial resistance may result in the actual soil constraint being less than the design value. This can lead to greater pipeline deformation and an increased risk of buckling failures (PRCI, 2009; Vazouras et al., 2021; Yoshizaki et al., 2003).

5. Conclusions

A large-scale experimental system was employed to investigate the axial behavior of steel pipes buried in unsaturated CDG. This study presents and analyses six tests with average matric suctions around the pipe ranging from 0 to 70.2 kPa, under nominal vertical pressures at the pipe center of 17 and 34 kPa. As far as the authors know, this is the first study examining the axial interaction between pipelines and unsaturated soil with fine particles. The experimental findings led to several key conclusions:

(a) The axial pullout resistance of the pipe increases as the matric suction increases from 0 to 70.2 kPa. Under the nominal pressure of 17 kPa, the axial resistance at an average suction of 70.2 kPa was 1.69 times greater than at 0 kPa (saturated condition). It highlights a significant underestimation of the load from soils to pipes in current design guidelines that are based on saturated soil mechanics, posing risks to pipeline systems.

(b) The additional interface contact pressure induced by capillary forces of liquid menisci between pipes and soils, $\chi_{int}s$, plays a significant role in axial resistance under unsaturated conditions. Of the suction-induced axial resistance increase at 70.2 kPa, 68 % is attributed to the increment in $\chi_{int}s$. The remaining part results from variations in average net interface contact pressure. During the pullout process, the constrained dilation plays a key role in increasing net interface contact pressure. The dilatancy and stiffness of CDG, two key factors of constrained dilation, increase as matric suction rises in CDG. Consequently, the increment in axial resistance due to net interface contact pressure variation rises from 10 % to 32 % as the suction increases from 17.8 to 70.2 kPa.

(c) A new simple model based on Bishop's stress was developed for calculating the axial pullout resistance in unsaturated soils. This method takes into account the additional interface contact pressure generated by capillary forces from liquid menisci between the pipes and soil, as well as the increased interface contact pressure resulting from the strongly constrained dilatancy of unsaturated soils. Compared to the existing equations in the design guidelines, the new equation demonstrated better performances when soils surrounding pipes are unsaturated.

CRediT authorship contribution statement

Chang Guo: Writing – original draft, Validation, Investigation, Visualization, Methodology, Formal analysis. **Chao Zhou:** Writing –

review & editing, Resources, Funding acquisition, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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